Elías Masquil

Detection Theory - Homework 1

Birthdays in a class

C_n : number of n-tuples of students having the same birthday

$P_n : P(C_n \geq 1)$

p_n : prob. at least one n-tuple and no (n+1)-tuple

1) Prove $P_n = 1 - \sum_{i}^{n-1} p_i$ and $P_n = P_{n-1} - p_{n-1}$

a) $P_n = 1 - P(C_n = 0) = 1 - P(C_n = 0, C_{n-1} > 0) - P(C_n = 0, C_{n-1} = 0)$

$= 1 - p_{n-1} - P(C_{n-1} = 0, C_{n-2} > 0) - P(C_{n-1} = 0, C_{n-2} = 0)$

...

$P_n = 1 - \sum_{i=1}^{n-1} p_i - \underbrace{P(C_2 = 0, C_1 = 0)}_{P(C_1 = 0) = 0}$ $\square$

b) $P_n - P_{n-1} = \sum_{i}^{n-2} p_i - \sum_{i}^{n-1} = -p_{n-1}$ $\square$

2) Prove $\mathbb{E}[C_n] = \frac{1}{365^{n-1}} \binom{30}{n}$

Let $X_{\{i_j\}_n}$ random variable / $X_{\{i_j\}_n} = 1 \iff$ alumns $i_1 \dots i_n$ share their birthday

$C_n = \sum_{1 \leq i_1 < \dots < i_n \leq 30} X_{\{i_j\}_n} \Rightarrow \mathbb{E}[C_n] = \sum_{i_1 < i_2 < \dots i_n} \mathbb{E}[X_{\{i_j\}_n}]$

$= \sum P[X_{\{i_j\}_n}] = \binom{30}{n} P[X_{\{i_j\}_n}] = \binom{30}{n} \frac{1}{365^{n-1}}$ $\square$

3) Prove $P(C_2 = 0) = \frac{365 \times \dots \times 336}{365^{30}}$

let's call b_i : birthday of alumn i

$P(C_2 = 0) = P(b_2 \neq b_1) P(b_3 \neq b_2 \cap b_3 \neq b_1) \dots P(b_n \neq b_{n-1} \dots b_n \neq b_1)$

$= \frac{364}{365} \cdot \frac{363}{365} \cdots \frac{336}{365}$ $\square$

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$$P_2 = 1 - \underbrace{P(C_2=0)}_{\approx 0{,}294} \approx 0{,}706$$

4) Prove

$$P_2 = \frac{1}{365^{30}} \sum_{i=1}^{15} \frac{\prod_{j=1}^{i} \binom{32-j}{2}}{i!} \prod_{k=0}^{29-i} (365-k)$$

$$P_2 = \sum_{i=1}^{15} \underbrace{P(C_2=i, C_3=0)}_{\frac{\#\text{ favorable cases}}{\text{total cases}} = \frac{\#\text{ favorable cases}}{365^{30}}}$$

$$P_2 = \frac{1}{365^{30}} \sum_{i=1}^{15} \underbrace{\{\#\text{ cases such that there are } i \text{ couples sharing b-day and no triplet}\}}_{\text{cases}(i)}$$

$i=1$

$$\text{cases}(1) = \underbrace{\binom{30}{2}}_{\text{ways to choose a couple in the class}} \underbrace{365}_{\text{possible days for the couples birthday}} \underbrace{364 \cdots 337}_{\text{possible birthdays for the rest}} = \binom{30}{2} \prod_{k=0}^{28} (365-k)$$

$i=2$

$$\text{cases}(2) = \frac{\underbrace{\binom{30}{2}}_{\text{ways to choose the first couple}} \underbrace{\binom{28}{2}}_{\text{ways to choose the second couple after the first}}}{\underbrace{2}_{\text{because we're counting duplicated examples}}} \underbrace{365}_{\text{possible days cople 1}} \underbrace{364}_{\text{possible days couple 2}} \underbrace{363 \cdots 338}_{\text{possible days for the rest}}$$

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Generalizing the formula for $i = 1, \dots, 15$

$$\text{cases}(i) = \underbrace{\frac{\prod_{j=1}^{i} \binom{30-2j}{2}}{i!}}_{\text{possible } i \text{ couples}} \underbrace{\prod_{k=0}^{29-i} (365-k)}_{\text{possible days combinations}}$$

$$\Rightarrow p_2 = \frac{1}{365^{30}} \sum_{i}^{15} \frac{\prod_{j=1}^{i} \binom{30-2j}{2}}{i!} \prod_{k=0}^{29-i} (365-k)$$

5) Computed in Python: $p_2 \approx 0{,}678$

6) $P_3 = 1 - \sum_{i=1}^{2} p_i = 1 - p_1 - p_2$

$p_1 = \mathbb{P}(C_2 = 0)$

$\Rightarrow \mathbb{P}_3 \approx 0{,}028$

7) Prove

$$p_3 = \frac{1}{365^{30}} \sum_{i=1}^{10} \frac{\prod_{j=1}^{i} \binom{33-3j}{3}}{i!} \left[\prod_{k=0}^{29-2i} (365-k) + \sum_{l=1}^{\left[\frac{30-3i}{2}\right]} \frac{\prod_{m=1}^{l} \binom{30-3i+2-2m}{2}}{l!} \prod_{n=0}^{29-2i-l} (365-n) \right]$$

$$P_3 = \sum_{i=1}^{10} \mathbb{P}(C_3 = i, C_4 = 0)$$

$$\mathbb{P}(C_3 = i) = \frac{\left(\begin{array}{c}\#\text{ possible different}\\ \text{triplets}\end{array}\right) \left[\left(\begin{array}{l}\#\text{ different bday configurations}\\ \text{when the rest of bdays are unique}\end{array}\right) + \left(\begin{array}{l}\#\text{ bday configurations when there are}\\ \text{also couples sharing bdays}\end{array}\right) \right]}{\#\text{ total birthdays}}$$

- \# total birthdays $= 365^{30}$
- \# possible triplets $= \dfrac{\prod_{j=1}^{i} \binom{33-3j}{3}}{i!}$

- # cases when the rest of bdays are unique

$$\underbrace{365 \cdot 364 \cdots}_{i\text{ - triplets birthdays}} \underbrace{\cdots (364 - i - (30-3i))}_{\text{rest of birthdays}}$$

$$= \prod_{k=0}^{29-2i} (365 - k)$$

- # cases when couples are allowed (it's analogous to the previous part)

We're going to sum the different cases from $l = 1$ to $\left[\frac{30-3i}{2}\right]$ couples sharing their birthday

cases with i triplets and l couples =

$$\underbrace{\frac{\prod_{m=1}^{l} \binom{30-3i+2-2m}{2}}{l!}}_{\text{possible couples}} \underbrace{\prod_{n=0}^{29-2i-l} (365-n)}_{\text{possible birthdays}}$$

Combining all the previous results we get the proposed formula

8) Computed in Python

$P_3 \approx 0{,}027998$ $\qquad$ $P_4 \approx 0{,}00054$

9)

$$P_n = \frac{1}{365^{30}} \sum_{i=1}^{\left[\frac{30}{n}\right]} \frac{\prod_{j=1}^{i} \binom{30+n-nj}{n}}{i!} \left[\prod_{k=0}^{29-(n-1)i} (365-k) + \sum_{s=2}^{n-1} \sum_{l=1}^{\left[\frac{30-ni}{s}\right]} \frac{\prod_{m=1}^{l} \binom{30-ni+s-sm}{s}}{l!} \prod_{t=0}^{29-si-l(s-1)} (365-t) \right]$$

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10) • $\mathbb{P}_{30} = \mathbb{P}(C_{30} = 1) = \frac{1}{365^{29}}$

$$\mathbb{E}[C_{30}] = \binom{30}{30} \frac{1}{365^{29}} = \mathbb{P}_{30}$$

• $\mathbb{E}[C_{29}] = \binom{30}{29} \frac{1}{365^{28}} = \frac{30}{365^{28}}$

$$\mathbb{P}_{29} = \mathbb{P}_{30} + \underbrace{\mathbb{P}(C_{29} = 1, C_{30} = 0)}_{\frac{\binom{30}{29} 365 . 364}{365^{30}}}$$

$$\mathbb{P}_{29} = \frac{1}{365^{29}} + \frac{30 . 364}{365^{29}}$$

11) Checked

12) Markov inequality: $\mathbb{P}_n \leq \mathbb{E}[C_n] \quad \forall n$

As n grows, $\mathbb{E}[C_n]$ decreases. Once that $\mathbb{E}[C_n]$ gets sufficiently small, and being smaller than 1, we have better estimates of $\mathbb{P}_n$.